Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset contains **167,826 rows** and **9 columns**, focusing on HIV epidemiology in children and adolescents. Here's a quick breakdown of its key characteristics:

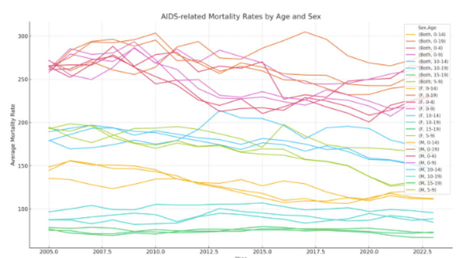
Key Columns:

- **Country/Region**: Name of the country or region.
- **UNICEF_Region**: Regional classification (some missing values).
- **Indicator**: The HIV-related measure (e.g., incidence rate, number of new infections).
- **Year**: The year of the estimate (float type).
- **Sex**: Sex category (Both, M, F).
- **Age**: Age group (e.g., 0–14, 15–19).
- **Value**, **Lower**, **Upper**: The estimate and its confidence bounds. These are currently strings and contain symbols like "<", so they'll need to be cleaned for numerical analysis.

Pl... :

2. Extract insights.
3. Deliver a concise, clear answer along with the exact Python code used. [>-]

What patterns are observed in AIDS-related mortality rates across different age and gender demographics?



Observed Patterns in AIDS-Related Mortality Rates:

1. **Higher Mortality in Younger Age Groups (0–14)**:
   - Across both sexes, the 0–14 age group consistently shows higher average AIDS-related mortality rates than the 15–19 group.

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2. ... but **females aged 15–19** show slightly higher mortality rates than their male counterparts in recent years.
   - In the 0–14 group, **gender disparities are minimal**, though slight variations occur over time.
3. **Temporal Trends**:
   - A general **decline in mortality rates over the years** is observed across all groups, reflecting global improvements in treatment and prevention.

Python Code Used:

python Copiar Editar

```python
# Filter dataset for AIDS-related mortality indicators
mortality_df = df[df['Indicator'].str.contains("AIDS-related deaths", case=False, na=Fals

# Clean and convert the 'Value' column to numeric
def clean_value(val):
    if isinstance(val, str):
        val = val.replace('<', '')
    try:
        return float(val)
    except ValueError:
        return None

mortality_df['Value'] = mortality_df['Value'].apply(clean_value)

# Group by Year, Sex, and Age
mortality_summary = (
    mortality_df
    .groupby(['Year', 'Sex', 'Age'])['Value']
    .mean()
    .reset_index()
    .pivot_table(index=['Year'], columns=['Sex', 'Age'], values='Value')
)
```

```
# Plotting
import matplotlib.pyplot as plt
mortality_summary.plot(figsize=(14, 8), title='AIDS-related Mortality Rates by Age and Se
plt.ylabel('Average Mortality Rate')
plt.xlabel('Year')
plt.grid(True)
plt.tight_layout()
plt.show()
```

Let me know if you'd like to zoom in on a specific region, country, or timeframe. [›_]